Anoop Kunchukuttan

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Research Interests

I am broadly interested in Natural Language Processing and Machine Learning. **Multilingual NLP and LLMs are my major research interests with a goal to investigate techniques for making quality NLP solutions available to multiple languages economically, efficiently and at scale.** This is important to make available the benefits of AI to large sections of non-English speaking population. In this context, I am interested in joint multilingual learning of LLM/NLP models, machine translation/ transliteration, cross-lingual NLP tasks, multilingual distributed representations, language typology and code mixing. To investigate these problems, I am generally interested in LLMs, multi-task learning, subword level models in NLP and unsupervised learning in NLP. I am also interested in exploring word and sentence representations as well as techniques for mining datasets at scale from large-scale corpora.

Data is a critical aspect to improve multilingual models. I am interested in **mining multilingual datasets at scale**, synthetic data generation, creating high-quality datasets and **evaluation benchmarks**. Recently, I have also been interested in reasoning models and agents (including multilingual settings) as well as speech LLMs.

I am very interested in looking at these problems **in the context of related languages, particularly Indian languages; and building open-source models/datasets/software for Indian language NLP.** To develop models for sovereign settings and serving solutions at population scale, I am interested in **efficient training and inference methods**.

Areas of Expertise

NLP: LLM Continual Pre-training, Reasoning & Post-training, Multilingual Learning, NLP for Indian languages, Self-supervised Learning, Machine Translation & Machine Transliteration, NLP for Related languages & low-resource languages, Distributed Representations for NLP, Large-scale Mining of NLP datasets.

Machine Learning: Self-supervised learning, supervised Classification, Unsupervised learning, Imbalanced dataset classification, optimizing performance metrics, prediction, sequence labelling, word alignment, string transduction and translation, sequence to sequence learning.

Education

- Ph.D in Computer Science & Engineering, *IIT Bombay*. 2012-2018.
Advisor: Prof. Pushpak Bhattacharyya.
Thesis: Machine Translation & Transliteration involving Related, Low-resource Languages

- M.Tech in Computer Science & Engineering, *IIT Bombay*. 2006-2008. CPI: 9.21
Advisor: Prof. Om Damani and Prof. Pushpak Bhattacharyya.
Thesis: Compound Noun Multiword Expression Extraction

- B.E in Computer Engineering, *University of Pune*. 2000-2004. % Marks: 65.53

Work Experience

Vice President (Research), Bodhan AI (*May 2026 - present*)

I lead the research team at Bodhan AI (formally the Centre of Excellence in AI for Education at IIT Madras) as Vice-President (Research). We are conducting foundational AI research to build digital public infrastructure for India's education system (the Bharat EduAI Stack). The research team works on fundamental AI particularly:

- Educational language models and other AI approaches that can help teachers teach better and students learn better by aligning language models to pedagogical and learning principles.
- Multilingual and multimodal AI to address India's linguistic diversity and serve multimodal experiences for providing high-quality AI-enabled education and learning.

Principal Applied Researcher, Microsoft (*Feb 2018 - April 2026*)

I have been part of the Machine Translation, Speech, Model Router and Azure AI Core teams.

- Machine Translation Team: Research and development of Neural Machine Translation & Transliteration for Indian languages and low-resource languages leading to shipping support for ~ 30 new languages while improving models for ~ 35 languages in Azure Translation Services. These were driven by contributions and innovations in the area of multilingual modeling, mining translation and transliteration datasets at web-scale, training recipes for multilingual modeling and low-resource languages.
- Speech: Research, development of Speech LLMs for Indian languages to improve Azure ASR models - current focus on improving base models for Indian languages and making them speech-aware via continued pre-training.
- Model Router: Research, development of Azure LLM Router for routing input queries to one LLM from a pool while balancing cost and accuracy. Contributed to adding support for new LLMs, building scalable training data pipelines and scaling the router to support a large number of models.
- Azure AI Core: (1) Enable inference and finetuning of open-source LLMs on internal, proprietary training infra by bridging the gaps between the OSS and internal frameworks, (2) Reinforcement Learning and finetuning for Diverse Domains.
- Mentoring data scientists on the team, driving learning initiatives.
- Driving external research collaborations.

Co-founder, Research Mentor, AI4Bharat, IIT Madras (*Jul 2019 - present*)
I am a founding and core member of the team building open-source datasets and models for Indian language NLP - leading research teams and mentoring students. In this role, I have also served as an **Adjunct Faculty** in the Department of Computer Science, **IIT Madras**.

- Contributions include widely used **models** (IndicTrans, IndicBERT, IndicBART, IndicWav2Vec, etc.), **datasets** (BPCC Corpus, Sangraha, IndicCorp, Aksharantar, ShrutiLipi, Naamapadam, etc.), and **benchmarks** (IndicGLEU, IndicXtreme, IndicSUPERB, etc.)
- These resources are widely used across the Indian NLP ecosystem and have significantly advanced Indian language NLP.

Teaching Assistant (Project), CFILT, IIT Bombay (*Jan 2012 - Dec 2017*)
Responsible for research and mentoring teams in collaborative projects with Xerox Research (crowdsourcing for collecting translation data), Crimson Interactive (grammatical error correction), Elsevier Publishing (medical domain translation).

Research Intern, Xerox Research Centre Europe (*Jul 2012 - Oct 2012*)
Mentors: Dr. Nicola Cancedda & Dr. Sriram Venkatapathy

Research Engineer, CFILT, IIT Bombay (*Mar 2011 - Dec 2011*)
Led the research and team working on a multi-stage crowdsourcing system for collecting translations for a complex domain like legal documents.

Module Lead, Persistent Systems (*Sep 2008 - Mar 2011*)
Research, development and team lead for search and information extraction from text documents in the medical domain in collaboration with academic universities in the US (Huntsman Cancer Institute, University of Washington at Saint Louis, BJC Hospitals). Using various ML models like SVM, CRF, Lucene, I worked on building the solutions listed below.

- High performance, scalable document de-identification (anonymization) and indexing system for free text medical records.

- A platform for building information extraction solutions for medical reports.
- Information extraction regarding Illicit Drug, Alcohol and Substance abuse history from free-text medical records.
- Information extraction from surgical pathology reports.

Member of Technical Staff, Persistent Systems (*Jul 2004 - Aug 2008*)
Software engineer working on medical device management and control software.

Awards & Honours

- KnowDis Machine Learning Award 2024 for contributions to Indian language NLP.
- Outstanding Paper Award at ACL 2024 conference for the IndicLLMSuite paper.
- Area Chair Award at ACL 2024 conference for the RomanSetu paper.
- NASSCOM AI Gamechangers Award 2022 for IndicWav2Vec (as part of AI4Bharat team).
- NASSCOM AI Gamechangers Award 2021 for Samanantar (as part of AI4Bharat team).
- Best Thesis Talk on *Investigations into subword units for Statistical Machine Translation between related languages* at Research and Innovation Symposium in Computing, IIT Bombay. 2017.
- Outstanding Paper Award. Workshop on Subword and Character level models in NLP. 2017.

Selected Publications

You can see the complete list and explore by research topics at `https://anoopkunchukuttan.github.io/publications`.
Journal and conference papers published at major venues like ACL, EMNLP, NAACL, COLING, AAAI, TACL, TMLR, ACM Computing Surveys.

Books

1. Anoop Kunchukuttan, Pushpak Bhattacharyya. *Machine Translation and Transliteration involving Related, Low-resource Languages*. CRC Press. 2021.

Journals

1. Sumanth Doddapaneni, Gowtham Ramesh, Mitesh M. Khapra, Anoop Kunchukuttan, Pratyush Kumar. *A Primer on Pretrained Multilingual Language Models*. **ACM Computing Surveys**. 2025.
2. Jay Gala, Pranjal A Chitale, Raghavan AK, Sumanth Doddapaneni, Varun Gumma, Aswanth Kumar, Janki Nawale, Anupama Sujatha, Ratish Puduppully, Vivek Raghavan, Pratyush Kumar, Mitesh M Khapra, Raj Dabre, Anoop Kunchukuttan. *IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages*. Transactions on Machine Learning Research **(TMLR)**. 2023.
3. Gowtham Ramesh, Sumanth Doddapaneni, Aravinth Bheemaraj, Mayank Jobanputra, Vivek Raghavan, Anoop Kunchukuttan, Pratyush Kumar, Mitesh Khapra and others. *Samanantar: The Largest Publicly Available Parallel Corpora Collection for 11 Indic Languages*. Transactions of the Association for Computational Linguistics **(TACL)**. 2022. *(NASSCOM AI Gamechangers Award)*
4. Raj Dabre, Chenhui Chu, Anoop Kunchukuttan. *A Comprehensive Survey of Multilingual Neural Machine Translation*. ACM Computing Surveys **(CSUR)**. 2020.
5. Pratik Jawanpuria, Arjun Balgovind, Anoop Kunchukuttan, Bamdev Mishra. *Learning Multilingual Word Embeddings in Latent Metric Space: A Geometric Approach*. Transactions of Association of Computational Linguistics **(TACL)**. 2019.
6. Anoop Kunchukuttan, Mitesh Khapra, Gurneet Singh, Pushpak Bhattacharyya. *Utilizing Orthographic Similarity for Multilingual Neural Machine Transliteration*. Transactions of the Association for Computational Linguistics **(TACL)**. 2018.

Conferences

1. Thanmay Jayakumar, Mohammed Safi Ur Rahman Khan, Raj Dabre, Ratish Puduppully, Anoop Kunchukuttan. *IndicIFEval: A Benchmark for Verifiable Instruction-Following Evaluation in 14 Indic Languages.* **AACL**. 2026.

2. Deepon Halder, Alan Saji, Thanmay Jayakumar, Ratish Puduppully, Anoop Kunchukuttan, Raj Dabre *RiddleBench: A New Generative Reasoning Benchmark for LLMs.* **EACL**. 2026.

3. Alan Saji, Raj Dabre, Anoop Kunchukuttan, Ratish Puduppully. *The Reasoning Lingua Franca: A Double-Edged Sword for Multilingual AI.* **EACL**. 2026.

4. Sumanth Doddapaneni, Mohammed Safi Ur Rahman Khan, Dilip Venkatesh, Raj Dabre, Anoop Kunchukuttan, Mitesh M Khapra. *Cross-Lingual Auto Evaluation for Assessing Multilingual LLMs.* **ACL**. 2025.

5. Sparsh Jain, Ashwin Sankar, Devilal Choudhary, Dhairya Suman, Nikhil Narasimhan, Mohammed Safi Ur Rahman Khan, Anoop Kunchukuttan, Mitesh M Khapra, Raj Dabre. *BhasaAnuvaad: A Speech Translation Dataset for 13 Indian Languages.* **ACL**. 2025.

6. Alan Saji, Jaavid Aktar Husain, Thanmay Jayakumar, Raj Dabre, Anoop Kunchukuttan, Ratish Puduppully. *RomanLens: The Role Of Latent Romanization In Multilinguality In LLMs.* **ACL Findings**. 2025.

7. Mohammed Safi Ur Rahman Khan, Priyam Mehta, Ananth Sankar, Umashankar Kumaravelan, Sumanth Doddapaneni, Sparsh Jain, Anoop Kunchukuttan, Pratyush Kumar, Raj Dabre, Mitesh M. Khapra. *IndicLLMSuite: A Blueprint for Creating Pre-training and Fine-Tuning Datasets for Indian Languages.* **ACL**. 2024. *(Outstanding Paper Award)*

8. Jaavid Aktar Husain, Raj Dabre, Aswanth Kumar, Ratish Puduppully, Anoop Kunchukuttan. *RomanSetu: Efficiently unlocking multilingual capabilities of Large Language Models models via Romanization.* **ACL**. 2024. *(Area Chair Award)*

9. Kaushal Maurya, Rahul Kejriwal, Maunendra Desarkar, Anoop Kunchukuttan. *CharSpan: Utilizing Lexical Similarity to Enable Zero-Shot Machine Translation for Extremely Low-resource Languages.* **EACL**. 2024.

10. Aswanth Kumar, Ratish Puduppully, Raj Dabre, Anoop Kunchukuttan. *In-context Example Selection for Machine Translation Using Multiple Features.* **EMNLP Findings**. 2023.

11. Yash Madhani, Sushane Parthan, Priyanka Bedekar, Gokul NC, Ruchi Khapra, Anoop Kunchukuttan, Pratyush Kumar, Mitesh M Khapra. *Aksharantar: Open Indic-language Transliteration datasets and models for the Next Billion Users.* **EMNLP Findings**. 2023.

12. Kaushal Bhogale, Abhigyan Raman, Tahir Javed, Sumanth Doddapaneni, Anoop Kunchukuttan, Pratyush Kumar, Mitesh M Khapra. *Effectiveness of mining audio and text pairs from public data for improving ASR systems for low-resource languages.* **ICASSP**. 2023.

13. Tahir Javed, Kaushal Bhogale, Abhigyan Raman, Pratyush Kumar, Anoop Kunchukuttan, and Mitesh M. Khapra. *IndicSUPERB: a speech processing universal performance benchmark for Indian languages.* **AAAI**. 2023.

14. Sumanth Doddapaneni, Rahul Aralikatte, Gowtham Ramesh, Shreya Goyal, Mitesh M Khapra, Anoop Kunchukuttan, Pratyush Kumar. *Towards Leaving No Indic Language Behind: Building Monolingual Corpora, Benchmark and Models for Indic Languages.* **ACL**. 2023.

15. Arnav Mhaske, Harshit Kedia, Sumanth Doddapaneni, Mitesh M Khapra, Pratyush Kumar, Rudra Murthy V, Anoop Kunchukuttan. *Naamapadam: A Large-Scale Named Entity Annotated Data for Indic Languages.* **ACL**. 2023.

16. Yash Madhani, Mitesh M Khapra, Anoop Kunchukuttan. *Bhasha-Abhijnaanam: Native-script and romanized Language Identification for 22 Indic languages.* **ACL**. 2023.

17. Ananya Sai B, Tanay Dixit, Vignesh Nagarajan, Anoop Kunchukuttan, Pratyush Kumar, Mitesh M Khapra, Raj Dabre. *IndicMT Eval: A Dataset to Meta-Evaluate Machine Translation Metrics for Indian Languages.* **ACL**. 2023.

18. Divyanshu Aggarwal, Vivek Gupta, Anoop Kunchukuttan. *IndicXNLI: Evaluating Multilingual Inference for Indian Languages.* **EMNLP**. 2022.

19. Aman Kumar, Himani Shrotriya, Prachi Sahu, Raj Dabre, Ratish Puduppully, Anoop Kunchukuttan, Amogh Mishra, Mitesh M Khapra, Pratyush Kumar. *IndicNLG Suite: Multilingual Datasets for Diverse NLG Tasks in Indic Languages*. **EMNLP**. 2022.

20. Raj Dabre, Himani Shrotriya, Anoop Kunchukuttan, Ratish Puduppully, Mitesh M. Khapra, Pratyush Kumar. *IndicBART: A Pre-trained Model for Natural Language Generation of Indic Languages*. Findings of the Association for Computational Linguistics: ACL 2022 **(ACL-Findings 2022)**. 2022.

21. Tahir Javed, Sumanth Doddapaneni, Abhigyan Raman, Kaushal Santosh Bhogale, Gowtham Ramesh, Anoop Kunchukuttan, Pratyush Kumar, Mitesh M. Khapra. *Towards Building ASR Systems for the Next Billion Users*. Proceedings of the AAAI Conference on Artificial Intelligence **(AAAI 2022)**. 2022. *(NASSCOM AI Gamechangers Award)*

22. Anoop Kunchukuttan, Siddharth Jain, Rahul Kejriwal.*A Large-scale Evaluation of Neural Machine Transliteration for Indic Languages*. Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics **(EACL 2021)**. 2021.

23. Divyanshu Kakwani, Anoop Kunchukuttan, Satish Golla, Gokul N.C., Avik Bhattacharyya, Mitesh M. Khapra, Pratyush Kumar. *IndicNLPSuite: Monolingual Corpora, Evaluation Benchmarks and Pre-trained Multilingual Language Models for Indian Languages*. Findings of EMNLP **(EMNLP-Findings**). 2020.

24. Anoop Kunchukuttan, Pratik Mehta, Pushpak Bhattacharyya. *The IIT Bombay English-Hindi Parallel Corpus*. Language Resources and Evaluation Conference **(LREC 2018)**. 2018.

25. Anoop Kunchukuttan, Maulik Shah, Pradyot Prakash, Pushpak Bhattacharyya. *Utilizing Lexical Similarity between Related, Low-resource Languages for Pivot-based SMT*. International Joint Conference on Natural Language Processing **(IJCNLP 2017)**. 2017.

26. Anoop Kunchukuttan, Pushpak Bhattacharyya. *Orthographic Syllable as basic unit for SMT between Related Languages*. Conference on Empirical Methods in Natural Language Processing **(EMNLP 2016)**. 2016.

27. Anoop Kunchukuttan, Mitesh Khapra, Pushpak Bhattacharyya. *Substring-based unsupervised transliteration with phonetic and contextual knowledge*. Conference on Natural Language Learning **(CoNLL 2016)**. 2016.

28. Anoop Kunchukuttan, Ratish Puduppully , Pushpak Bhattacharyya. *Brahmi-Net: A transliteration and script conversion system for languages of the Indian subcontinent*. Conference of the North American Chapter of the Association for Computational Linguistics - Human Language Technologies: System Demonstrations **(NAACL 2015)**. 2015.

29. Anoop Kunchukuttan, Abhijit Mishra, Rajen Chatterjee, Ritesh Shah, Pushpak Bhattacharyya. *Shata-Anuvadak: Tackling Multiway Translation of Indian Languages* . Language and Resources and Evaluation Conference **(LREC 2014)**. 2014.

30. Anoop Kunchukuttan, Rajen Chatterjee, Shourya Roy, Abhijit Mishra and Pushpak Bhattacharyya. *Trans-Doop: A Map-Reduce based Crowdsourced Translation for Complex Domain* . Proceedings of the Association of Computational Linguistics: System Demonstrations **(ACL 2013)**. 2013.

Selected Invited Talks & Teaching

The complete list of talks along with slides and videos is available at `https://anoopkunchukuttan.github.io/talks`

1. **Tutorial** on *Data and Model Centric Approaches for Expansion of Large Language Models to New languages* at **EMNLP**, Suzhou, China, in December 2025 with Raj Dabre, Rudramurthy V, Mohammed Safi Ur Rahman & Thanmay Jayakumar.

2. **Tutorial** on *Multilingual Language Modeling* at the Winter School on Deep Learning, **IIT Guwahati**, in Jan 2025.

3. **Tutorial** on *Multilingual Neural Machine Translation* at **COLING**, Barcelona, Spain, in September 2020 with Raj Dabre & Chenhui Chu.

4. **Tutorial** on *Building Multilingual NLP datasets at scale* at the IASNLP Summer School, IIIT Hyderabad, in June 2025.

5. **Tutorial** on *Statistical Machine Translation between related languages* at **North American Chapter of the Association for Computational Linguistics (NAACL)**, San Diego, United States, in June 2016 with Prof. Pushpak Bhattacharyya & Mitesh Khapra.
6. **Tutorial** on *the AI4Bharat Initiative* at the ICON 2021, in Dec 2021 with Mitesh Khapra and Pratyush Kumar.
7. **Tutorial** on *Machine Translation* at the ACM India Summer School on Natural Language Processing, IIIT Hyderabad, in July 2021.
8. **Podcast** on *Is the Artificial Intelligence boom a bubble?* at The Hindu In-Focus Parley with Dr. Bhagwan Chowdhry and Areena Arora. December 2025.
9. **Invited Talk** on *Indic NLP: A Multilinguality and Language Relatedness Perspective* at Vaibhav Summit (Organized by MyGov, Govt. of India). October 2020.
10. **Invited Talk** on *NLP for Indian Languages: A Language Relatedness Perspective* at NASSCOM Data Science & AI - Center of Excellence, Bengaluru in August 2019.
11. **Mini Course** on *Natural Language Processing - A Deep Learning Approach* at IIT Alumni Center Bengaluru AI Deep Dive Workshop, July 2019.
12. **Course lectures** on Deep Learning covering sequence learning, labeling, transformers and machine translation at IIT Bombay and IIT Hyderabad.

Academic Services and Activities

- Co-Founder and Co-Lead of *AI4Bharat*, a research group at IIT Madras working to advance Indian language NLP.
- Adjunct Faculty, Department of Computer Science, IIT Madras, Aug 2022 to July 2024.
- Area Chair, ACL Rolling Review (Dec 2023 to *present*).
- Member of Academic Council, IIIT Hyderabad (Apr 2025 to *present*).
- Member of CLD Program Curriculum and Review Committee, IIIT Hyderabad (Apr 2025 to *present*).
- Member of Textbook Development Team (TDT) for Grade 11 and 12 Textbooks in Foundations and Methods of Artificial Intelligence for NCERT (Dec 2026 to *present*).
- Member of Executive Council, Bhashini (Jan 2026 to *present*)
- Member - Technical Evaluation Committee, National e-Governance Division (March 2026 to *present*).
- Founder and Maintainer of *Indic NLP Catalog*, a catalog for indexing Indian language NLP resources.
- Journal Reviewer: Computational Linguistics (CL), Transactions of ACL (TACL), Transaction of Machine Learning Research (TMLR), ACM Transactions Asian & Low-Resource Language Information Processing (TALLIP), Natural Language Engineering (NLE).
- Conference Reviewer: ACL, EMNLP, NAACL, NeurIPS, EACL, COLING, ICLR, AAAI, IJCAI, COLM, WMT, LREC, MT Summit, EAMT, CODS-CoMAD, ACML, ICON.
- Demo Chair: CoDS-COMAD 2023.
- Co-organizer for Workshop on Asian Language Translation shared task (2016-2024).
- Organizing Committee and Workshop Chair, COLING 2012.
- Co-advising students at IIT Madras, IIT Bombay and IIT Hyderabad.
- Masters' and Ph.D theses reviews.

Software & Resources created

Source Code on **Github**: `https://github.com/anoopkunchukuttan`, `https://github.com/AI4Bharat`, `https://github.com/Bodhan-AI`
Data and Models on **HuggingFace**: `https://huggingface.co/ai4bharat`, `https://huggingface.co/bodhan-ai`

Software

- *Indic NLP Library*: NLP library for Indian languages covering normalizer, transliterator, word segmenter, script information phonetic similarity, syllabification, etc.
- *Models from AI4Bharat*: CIA-Hercules LLM Evaluator, Airavat LLM, IndicBERT, IndicBART, IndicTrans, IndicXlit, IndicWav2Vec, IndicFT.
- *GEOMM*: Geometry-Aware Multilingual Mapping, a toolkit for learning multilingual word embeddings. More generally, it can be used to learn mappings between different high dimensional spaces.
- IIT Bombay Unsupervised Transliterator: Unsupervised transliteration system which uses phonetic features to define transliteration priors. This is an EM based method which builds on Ravi and Knight's 2009 work.
- Multilingual Neural Machine Translation System: A multilingual Neural Machine Translation system written in Tensorflow.
- *METEOR-Indic*: MT evaluation tool extended for 18 Indian languages.
- *Moses Job Scripts*: A simple experiment management system for Moses.
- *CFILT Pre-order (Maintainer)*: Source-side pre-ordering of English for English to Indian language translation.
- *McTorch*: A manifold optimization library for deep learning.

Resources

- *Reasoning Model Training Demo*: Hands-on tutorial from our team to train reasoning models at scale with open-source software and leveraging the power of Azure ML to make the process easy at Microsoft Build 2025.
- *Datasets from AI4Bharat*: BhasaAnuvaad, RECON, Pralekha, IndicLLMSuite, IndicXtreme, IndicLID, IndicMT-Eval, IndicCorp, Samanantar, Aksharantar, IndicGLUE, IndicNLGSuite, ShrutiLipi, IndicWav2Vec, IndicFT.
- IIT Bombay English-Hindi Parallel Corpus
- Indian Language NLP Resources Catalog
- Mined Transliteration corpora for 110 Indian language pairs
- Transliteration corpora for English-Hindi gathered through crowdsourcing
- GEOMM Multilingual Word Embeddings for Indian languages
- Translation Resources for 110 Indian language pairs

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